Random Assignment Processes

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Abstract

In the framework of random assignment processes, which randomly assign an index, within a finite set of labels, to the points of an arbitrary set, we propose sufficient conditions for a family of finite-valued nonexchangeable random variables to verify a De Finetti theorem, meaning that they are conditionally independent given some other random variable. By the way, the analysis of random assignments process gives some interesting strong laws of large numbers.

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1 Introduction

In the probability literature, a De Finetti theorem usually refers to a result ensuring that some random variables are conditionally independent given some other random variable. In this sense, the classical De Finetti theorem states that an infinite number of exchangeable (and, in particular, identically distributed) random variables taking values in $\{0, 1\}$ are conditionally independent (and so, they are conditionally i.i.d.); see, e.g., [3, 4].

Generalizations of this "first" De Finetti theorem were aimed towards showing that exchangeable random variables taking values in domains larger that $\{0, 1\}$ are conditionally independent as well; see, e.g., [1, Chapter 7.3], [4, Chapter 27], [5], or [6, Chapter 8]. Our recent research, however, has focused on obtaining De Finetti theorems when *relaxing the exchangeability assumption*.

In this sense, sufficient conditions for a family $\{Y(x)\}_{x\in E}$ of non identically distributed $\{0, 1\}$-valued random variables to verify a De Finetti theorem are proposed in [10, 12] by introducing the concept of a (weighted) random selection process, with $Y(x) = 1$ when the element $x \in E$ is "selected", while $Y(x) = 0$ if $x \in E$ is "rejected". Then, under adequate hypotheses, the random variables $\{Y(x)\}_{x\in E}$ are shown to be conditionally independent, but neither i.d. nor exchangeable. In [11] we have proved that these hypotheses are also necessary to get a De Finetti theorem for such random selection processes.

In [9], random assignment processes has been proposed as a generalization of random selection processes. The random variables $\{Y(x)\}_{x\in E}$ are now supposed to take values in a finite set $L = \{0, 1, \ldots, \ell\}$. So, the value of $Y(x)$ can be interpreted as the label or index assigned to the point $x \in E$. The labels are assumed to be randomly assigned according to a multidimensional weight function (the details are given in Section 2 below). Then, under further appropriate conditions, it is shown that the L-valued random variables $\{Y(x)\}_{x\in E}$ satisfy a De Finetti theorem, that is, they are con-

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ditionally independent though, once again, they are neither i.d. nor exchangeable. Now, the sufficient conditions for a De Finetti theorem proposed in [9] are widely generalized and, in particular, we give a satisfactory answer to the conjecture stated in the conclusions of [9].

2 Random assignment processes

Suppose that $(\Omega, \mathcal{F}, \mathrm{P})$ is a probability space, E is an arbitrary set, and

$$L = \{0, 1, 2, \ldots, \ell\}$$

for some positive integer ℓ is a set of indices.

We consider a family $\boldsymbol{Y} = \{Y(x)\}_{x \in E}$ of random variables

$$Y(x) : \Omega \to L \quad \text{for every } x \in E.$$

that assign to each point $x \in E$ an index in L. Later we will impose some conditions on the probability measure P so that $\boldsymbol{Y}$ has suitable properties.

First of all, we introduce some notation. Let $\{\boldsymbol{e}_i\}_{1 \le i \le \ell}$ be the canonical basis of $\mathbb{R}^\ell$, and let $\boldsymbol{e_0} = \boldsymbol{0} \in \mathbb{R}^\ell$. The random variables $Y(x)$ may be alternatively represented by means of the random vectors

$$N(x) = \boldsymbol{e}_{Y(x)} = [0, \ldots, 0, \underbrace{1}_{Y(x)}, 0, \ldots, 0].$$

More generally, if $\mathcal{B}$ is the ring of all finite subsets of E, we can define, for each $B \in \mathcal{B}$,

$$N(B) = \sum_{x \in B} N(x) = \big[N_1(B), \ldots, N_\ell(B)\big]$$

which takes values in

$$R(\#B) = \big\{\boldsymbol{r} \in \mathbb{N}^\ell : \bar{r} \le \#B\big\},$$

where $\#B$ denotes the cardinal of B and $\bar{r} = \sum_{i=1}^{\ell} r_i$ is the sum of the components of $\boldsymbol{r} \in \mathbb{N}^\ell$. Equivalently, $N(B) = \boldsymbol{r}$ means that there are $N_i(B) = r_i$ points in B with $Y(x) = i$, for $1 \le i \le \ell$, and

$$\#B - \sum_{i=1}^{\ell} N_i(B) = N_0(B) = r_0 = \#B - \bar{r}$$

points in B with $Y(x) = 0$.

Invariant σ-algebras.

Given a finite set $B \in \mathcal{B}$, let $\mathcal{G}_B$ be the sub-σ-algebra of $\mathcal{F}$ consisting of all the events of the form $\{\boldsymbol{Y} \in F\}$, where F is a measurable subset of L^E which is invariant under any permutation of the elements of B. It is easy to show that $\mathcal{G}_B$ is the σ-algebra generated by $N(B)$ and the $Y(x)$ for $x \notin B$.

For an arbitrary subset $A \subset E$, we define the σ-algebra of events invariant under any finite permutation in A as

$$\mathcal{G}_A = \bigcap_{B \in \mathcal{B}, B \subset A} \mathcal{G}_B.$$

Obviously, if $A_1 \subset A_2$ then $\mathcal{G}_{A_2} \subset \mathcal{G}_{A_1}$.

Definition of a random assignment process. Let $\boldsymbol{m} : E \to (0, \infty)^\ell$ be a function that associates a vector of positive weights

$$\boldsymbol{m}(x) = \big[m_1(x), \ldots, m_\ell(x)\big]$$

to each $x \in E$.

Definition 2.1 *A family of L-valued random variables $\boldsymbol{Y} = \{Y(x)\}_{x \in E}$ on $(\Omega, \mathcal{F}, \mathrm{P})$ or as well the family $\boldsymbol{N} = \{N(B)\}_{B \in \mathcal{B}}$, is a random assignment process on $(E, \boldsymbol{m})$ if the following conditions hold for all $B \in \mathcal{B}$:*

(C1) *Conditional on $N(B)$, $\{Y(x)\}_{x \in B}$ and $\{Y(x)\}_{x \in B^c}$ are independent.*

(C2) *For all $i = 1, \ldots \ell$ and $x \in B$,*

$$\mathrm{P}\{Y(x) = i \mid N(B) = \boldsymbol{e}_i\} = \frac{m_i(x)}{m_i(B)}.$$

where $m_i(B) = \sum_{x \in B} m_i(x)$.

The Markovian condition (C1) means that, once the numbers of the different indices attributed to the points in B are fixed, their positions inside B do not depend on the indices attributed to the points outside B. On the other hand, the condition (C2) establishes that, if a unique positive index $i \in L$ is to be located in B, then this positive index corresponds to $x \in B$

with probability proportional to its weight $m_i(x)$.

Conditional distributions. It is easy to prove [9, Proposition 2.1] that, for each $B \in \mathcal{B}$, $\boldsymbol{r} \in R(\#B)$, and any disjoint sets $B_1, \ldots, B_\ell \subset B$ with $\#B_i = r_i$ for $i = 1, \ldots, \ell$, we have

$$\mathrm{P}\Big(\bigcap_{i=1}^{\ell}\{Y(x) = i \ \forall x \in B_i\} \mid N(B) = \boldsymbol{r}\Big) = \frac{1}{S(B,\boldsymbol{r})} \prod_{x\in B_1} m_1(x) \cdots \prod_{x \in B_l} m_\ell(x) \quad (2.1)$$

where the normalizing factor $S(B, \boldsymbol{r})$ is the sum of the products in the right hand side of (2.1) over all the disjoint subsets of B with $\#B_i = r_i$ for $i = 1, \ldots, \ell$. In case that $\boldsymbol{r} \notin R(\#B)$, we define $S(B, \boldsymbol{r}) = 0$.

Fix $B \in \mathcal{B}$ and a subset $C \subset B$. As a direct consequence of (2.1) (cf. [9, Proposition 2.3]), for any $\boldsymbol{r} \in R(\#B)$ and $\boldsymbol{k} \in R(\#C)$ such that $\boldsymbol{r} - \boldsymbol{k} \in R(\#B - \#C)$, the probability that $N(C) = \boldsymbol{k}$ conditional on $N(B) = \boldsymbol{r}$ is the following "generalized" multidimensional hypergeometric distribution

$$\mathrm{P}\{N(C) = \boldsymbol{k} \mid N(B) = \boldsymbol{r}\} = \frac{S(C,\boldsymbol{k})\, S(B - C, \boldsymbol{r} - \boldsymbol{k})}{S(B,\boldsymbol{r})}. \quad (2.2)$$

More generally, if $C_1, \ldots, C_s$ are disjoint subsets of $B \in \mathcal{B}$, then

$$\mathrm{P}\{N(C_1) = \boldsymbol{k}_1, \ldots, N(C_s) = \boldsymbol{k}_s \mid N(B) = \boldsymbol{r}\} = \frac{S(C_0, \boldsymbol{k}_0) S(C_1, \boldsymbol{k}_1) \cdots S(C_s, \boldsymbol{k}_s)}{S(B, \boldsymbol{r})} \quad (2.3)$$

with $C_0 = B - \cup_{t=1}^{s} C_t$ and $\boldsymbol{k}_0 = \boldsymbol{r} - \sum_{t=1}^{s} \boldsymbol{k}_t$.

Marginal distributions. The above results show that the distribution of the random assignment process $\boldsymbol{N}$ inside $B \in \mathcal{B}$ is determined by the marginal distribution of $N(B)$. Hence, the distribution of the random assignment process is completely determined by a family of *compatible marginal distributions* for all $N(B)$ with $B \in \mathcal{B}$, meaning that for every disjoint B and C in $\mathcal{B}$ the total probability rule

$$\mathrm{P}\{N(B) = \boldsymbol{k}\} = \sum_{\boldsymbol{r} \in R(\#C)} \mathrm{P}\{N(B \cup C) = \boldsymbol{k} + \boldsymbol{r}\} \times \frac{S(C, \boldsymbol{r}) S(B, \boldsymbol{k})}{S(B \cup C, \boldsymbol{k} + \boldsymbol{r})} \quad (2.4)$$

holds.

Clearly, if E is finite, then we can choose an arbitrary distribution for $N(E)$, and then derive the joint distributions of the random assignment process by using (2.3) with $B = E$. Therefore, to avoid this trivial situation, *in what follows we will assume that the set E is not finite.*

As shown in [9, Proposition 2.5], an obvious choice of compatible marginal distributions for any E and $\boldsymbol{m}$ is the following mixture of "generalized" multinomial distributions: for all $B \in \mathcal{B}$ and all $\boldsymbol{k} \in R(\#B)$,

$$\mathrm{P}\{N(B) = \boldsymbol{k}\} = \int_{\Delta_\ell} \frac{S(B, \boldsymbol{k})\ \xi_1^{k_1} \cdots \xi_\ell^{k_\ell} \xi_0^{\#B - \bar{k}}}{\prod_{x \in B} \boldsymbol{\xi} \cdot \boldsymbol{m}(x)}\ F(d\boldsymbol{\xi}) \quad (2.5)$$

where $\Delta_\ell = \{\boldsymbol{\xi} \in [0,1]^L : \boldsymbol{\xi} \cdot \boldsymbol{1} = 1\}$ and F is an arbitrary distribution on Δ_ℓ. We note that, in order to simplify some expressions, in (2.5) we let $m_0(x) = 1$ and

$$\boldsymbol{m}(x) = [m_0(x), m_1(x), \ldots, m_\ell(x)]$$

for all $x \in E$.

Remark 2.2 *In this setting, $0 \in L$ seems to play a particular role as a "default" index. However, we may suppose that we are given an arbitrary positive vector $\boldsymbol{m}(x) = [m_0(x), m_1(x), \ldots, m_\ell(x)]$ and then, replacing (C2) with*
(C2)' *For all $i, j \in L$ with $i \neq j$,*

$$\mathrm{P}\{Y(x) = i \mid N_i(B) = 1, N_j(B) = \#B - 1\} = \frac{m_i(x)/m_j(x)}{\sum_{y \in B} m_i(y)/m_j(y)}$$

would give identical results, with the factor $\prod_{x\in B_0} m_0(x)$ *added to the righthand side of (2.1).*

Thus, there is no loss of generality when letting $j=0$ *and* $m_0(x)\equiv 1$ *in* (C2)'. *This explains why the index* 0 *has no special relevance whereas, on the contrary, the quotients* $m_i(x)/m_j(x)$ *will be particularly relevant for our analysis.*

In case that (2.5) holds for some F, the random variables $\{Y(x)\}_{x\in E}$ are conditionally independent given $\boldsymbol{\xi}$, with conditional distributions

$$\mathrm{P}\{Y(x)=i\mid \boldsymbol{\xi}\} = \frac{\xi_i\, m_i(x)}{\boldsymbol{\xi}\cdot \boldsymbol{m}(x)} \quad \text{for } i\in L. \tag{2.6}$$

Therefore, *we say that a De Finetti-type result holds.*

Main results. Our principal goal is to give conditions on the weight function $\boldsymbol{m}$ ensuring that, given a random assignment process on $(E,\boldsymbol{m})$, the marginal distributions $N(B)$ for $B\in\mathcal{B}$ are *necessarily* as in (2.5) for some distribution F on Δ_ℓ (and so, a De Finetti-type theorem holds).

It was shown in [9, Theorem 3.5] that a De Finetti-type theorem holds if the following condition is satisfied:
(A) There is a sequence $A=\{x_n\}_{n\geq 1}\subset E$ such that

$$0<\lim_{n\to\infty} m_i(x_n)<\infty$$

for every $1\leq i\leq \ell$.

Letting $A_n=\{x_q\}_{1\leq q\leq n}$, this condition implies that the $m_i(A_n)$ grow to infinity, as $n\to\infty$, at the same speed (proportional to n) for all $i\in L$.

Correspondingly, we prove in [9, Theorem 3.4] that

$$\left[\frac{N_1(A_n)}{n},\ldots,\frac{N_\ell(A_n)}{n}\right]\xrightarrow{\text{a.s.}}\boldsymbol{\xi} \tag{2.7}$$

where $\boldsymbol{\xi}\in\Delta_\ell$ is a random vector.

Here, we improve [9, Theorem 3.5] by replacing (A) with the weaker condition (A*) below, in which, for $x\in E$ and $i,j\in L$, we denote $m_{j|i}(x)=m_j(x)/m_i(x)$:
(B) There is a sequence $A=\{x_n\}_{n\geq 1}\subset E$ such that, for every i and j in L, the limit $\lim_n m_{j|i}(x_n)=m_{j|i}\in[0,\infty]$ exists and, moreover,

$$m_{j|i}(A)=\sum_{n=1}^{\infty} m_{j|i}(x_n)=\infty.$$

We note that, regardless the condition (B) holds, there is indeed a sequence $A\subset E$ such that the limits $m_{j|i}$ exist (just choose a suitable subsequence of any given sequence in E). Besides, the series $m_{j|i}(A)$ diverges whenever $m_{j|i}>0$. Hence, the condition (B) just imposes the existence of a sequence A such that $m_{j|i}(A)=\infty$ even if $m_{j|i}=0$.

When (A) is replaced by (B), the strong law of large numbers (2.7) has a much more involved form. Basically, if for each $M\subset L$, let be

$$\Omega_M=\big\{N_i(A)>0\ \ \forall i\in M\big\}\cap \big\{N_j(A)=0\ \ \forall j\notin M\big\}$$

then

$$\Omega=\textstyle\bigcup_{\mathrm{P}(\Omega_M)>0}\Omega_M\cup N$$

with $\mathrm{P}(N)=0$ and, furthermore, for each M with $\mathrm{P}(\Omega_M)>0$, there exists a set $C_0\subset M$ such that, $\mathrm{P}(\cdot\mid\Omega_M)$-a.s.

$$\frac{N_i(A_n)}{n}\longrightarrow\frac{\xi_i}{\alpha_0}\ \ \forall i\in C_0$$
$$\frac{N_j(A_n)}{m_{j|i}(A_n)}\longrightarrow\frac{\xi_j}{\alpha_0}\ \ \forall j\notin C_0,$$

where $\alpha_0=\sum_{i\in C_0}\xi_i$.

This strong law of large numbers allows to get the following De Finetti's result:

Theorem 2.3 *Let* $\boldsymbol{Y}$ *be a random assignment process on* $(E,\boldsymbol{m})$, *and suppose that the condition (B) is verified for some sequence* $A\subset E$. *Then there exists a* $\mathcal{G}_A$*-measurable random vector* $\boldsymbol{\xi}$ *with values in* Δ_ℓ *such that* $\{Y(x)\}_{x\in E}$ *are conditionally independent given* $\boldsymbol{\xi}$ *and (2.6) holds, that is,*

$$\mathrm{P}\{Y(x)=i\mid\boldsymbol{\xi}\}=\frac{\xi_i\, m_i(x)}{\boldsymbol{\xi}\cdot\boldsymbol{m}(x)}$$

for all $x \in E$ and $i \in L$. Consequently, for all $B \in \mathcal{B}$ and $\boldsymbol{k} \in R(\#B)$,

$$\mathrm{P}\{N(B) = \boldsymbol{k}\} = \mathrm{E}\left[\frac{S(B,\boldsymbol{k})\ \xi_1^{k_1} \cdots \xi_\ell^{k_\ell} \xi_0^{\#B-\bar{k}}}{\prod_{x\in B} \boldsymbol{\xi} \cdot \boldsymbol{m}(x)}\right]. \tag{2.8}$$

This results generalizes the classical De Finetti theorem for sequences of exchangeable finite-valued random variables. Indeed, it is easy to prove that the L-valued random variables $\{Y(x)\}_{x\in E}$ are exchangeable if and only if $\{Y(x)\}_{x\in E}$ is a random assignment process on $(E, \boldsymbol{m})$, where $\boldsymbol{m}$ is the constant weight function $m_i(x) = 1$ for all $i \in L$ and $x \in E$ (cf. Remark 2.4 in [9]).

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